

Dewaner pada
on 19.09

~2.

Demo:

$$m = 1 \text{ kg}$$

$$x(t) = 5 \cos(5t)$$

$$t = 20$$

$$|F| = ?$$

Terence:

$$V = \frac{dx(t)}{dt} = -25 \sin 5t$$

$$a = \frac{dV(t)}{dt} = -125 \cos 5t$$

$$F = ma = -125 \cos 10 \text{ N}$$

$$\text{Answer: } 125 \cos 10 \text{ N}$$

~3.

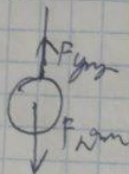
Demo:

$$m = 1 \text{ kg}$$

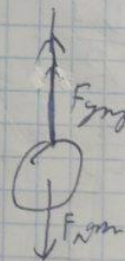
$$a = 5 \frac{\text{m}}{\text{s}^2}$$

$$F_{\text{mg}} = ?$$

Terence:



=>



$$a = 5 \frac{\text{m}}{\text{s}^2}$$

$$F_{\text{mg}} - F_{\text{nm}} = ma$$

$$F_{\text{mg}} = m(g + a) = 14,8 \text{ N}$$

$$\text{Answer: } 14,8 \text{ N}$$

Demo:

$$m = 50000$$

$$F_{\text{mg}} = 9800$$

$$t = 600$$

$$V_1 = 20$$

$$V_0 = ?$$

Answer: 11

Demo:

$$V = 0$$

$$L = 300$$

$$F_{\text{mg}} = ?$$

Given:

$$m = 500000 \text{ kg}$$

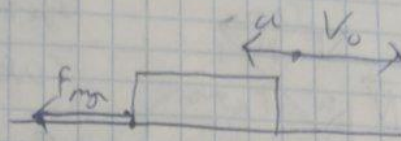
$$F_{\text{fr}} = 980000 \text{ N}$$

$$t = 600$$

$$V_1 = 20$$

$$V_0 = ?$$

~ 4.
Derive:



$$F_{\text{fr}} = \mu N \quad (\text{II given})$$

$$\Rightarrow a = -\frac{980000}{500000} = -\frac{98}{500}$$

$$= -\frac{196}{1000} = -0,196 \frac{\text{m}}{\text{s}^2}$$

$$V_1 = V_0 - at; \quad V_0 = V_1 + at =$$

$$= 1 - 0,196 \cdot 60 = 1,96 \cdot 6 = 11,76 \frac{\text{m}}{\text{s}}$$

$$\text{Answer: } 11,76 \frac{\text{m}}{\text{s}}$$

~ 5.

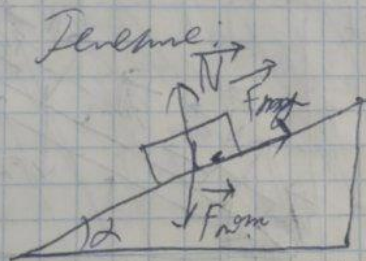
Given:

$$V = 0$$

$$\alpha = 30^\circ$$

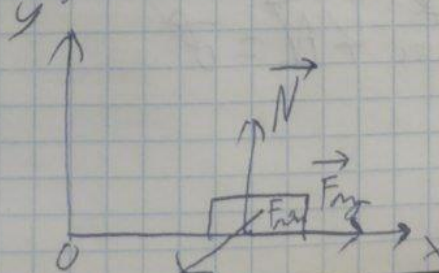
$$F_{\text{fr}} = ?$$

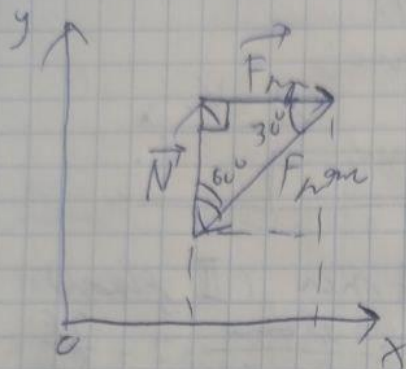
Derive:



$$\text{no } \vec{N} + \vec{F}_{\text{fr}} + \vec{F}_{\text{grav}} = \vec{0}$$

balans och kejsplanen:





$\cos \alpha = \frac{F_{rx}}{F_{g \sin \alpha}}$
 $F_{rx} = \cos \alpha \cdot F_{g \sin \alpha}$
 $\tan \alpha = \frac{N_x}{F_{rx}}$

$F_{rx} = |F_{g \sin \alpha}| \cos \alpha$

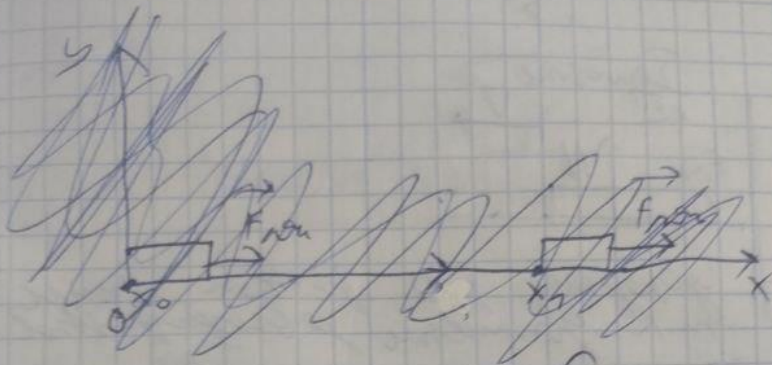
$F_{rx} = \cos^2 \alpha \cdot mg = \frac{3}{4} \cdot \frac{5}{2} = 2.5 \text{ N}$

Orbit: 2,5 N (N)

Demo:
 $v = \text{const}$
~~no I~~



no I zeroing:
 $F_{rx} + F_{ry} + N = 0$



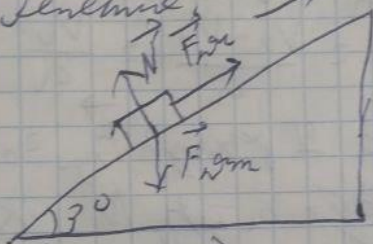
Demo:

$$V = 0,1 \text{ m/s}$$

$$\alpha = 30^\circ$$

$$|F_{fr}| = ?$$

Jenelme:



$$F_{fr} = F_{grav} \cos \alpha$$

$$F_{fr} \times = m g \cos \alpha$$

$$F_{fr} \times = |F_{fr}| = \cos^2 \alpha m g = 2,5 \text{ m}$$

10 I geyang:

$$\vec{F}_{fr} + \vec{F}_{grav} + \vec{N} = \vec{0}$$

(m.k. $V = \text{const}$)

Orbani 2,5 m

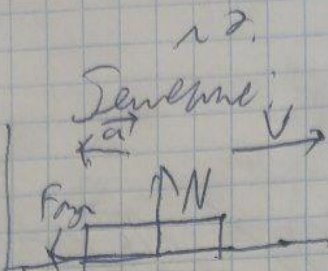
Dado:

$$V = V_0$$

$$V_1 = 0$$

$$k = k$$

t = ?



Yoga para o corpo humano
m.k. gromulme perogromulme

$$V_1 = V_0 - at$$

$$V_1 - V_0 = -at$$

$$t = \frac{V_0 - V_1}{a} = \frac{V_0}{a}$$

(ho II zekony) $F_{mg} = ma = kN$

$$a = \frac{kN}{m} = \frac{kN}{m}$$

(ho III zekony) $N = F_{mg} = mg$

$$a = \frac{kmg}{m} = kg$$

zayade: $t = \frac{V_0}{a} = \frac{V_0}{kg}$ c

Orben: $t = \frac{V_0}{kg}$ c

Dans:

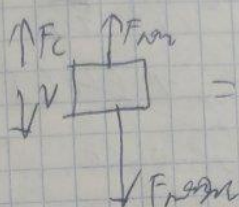
$$m_0 = 1000 \text{ kg}$$

$$F_{\text{non}} = 12000 \text{ N}$$

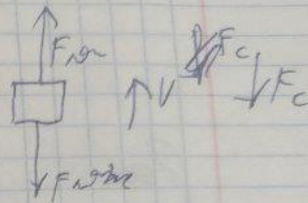
$$m_1 = ?$$

et.

Données:



=>



1) (no II zéro) $F_c + F_{\text{non}} + F_{\text{grav}} = 0$; $F_c = m_0 g - F_{\text{non}}$

2) (no II zéro) $F_{\text{non}} = F_{\text{grav}}$; $F_{\text{grav}} = m_1 g = F_c$

$$\Rightarrow m_1 g = m_0 g - F_{\text{non}}$$

$$m_1 = m_0 - \frac{F_{\text{non}}}{g} = \left| 1000 - \frac{12000}{10} \right| = 400 \text{ kg}$$

Donc: 400 kg

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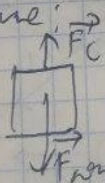
Dans:

$$m = 2 \text{ kg}$$

$$a = 5 \frac{\text{m}}{\text{s}^2}$$

$$F_c$$

Données:



$$a$$

(II zéro) $F_{\text{non}} = m g = 20 \text{ N}$

$$\Delta a = g - a = 5 \frac{\text{m}}{\text{s}^2}; |F_c| = 2 \cdot 5 = 10 \text{ N}$$

Donc: 10 N